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Electronic device including boost circuit

Abstract

An electronic device including a boost circuit includes a first boost circuit, a first output circuit, a load, a first storage capacitor, a second storage capacitor, and an input terminal, wherein the input terminal is connected to the first storage capacitor and the first boost circuit, the second storage capacitor is connected to the first boost circuit and the first output circuit, and the load is connected to the first output circuit.

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Background/Summary

RELATED APPLICATIONS
(1) This application claims priority to Japanese Patent Application No. 2021-041967, filed on Mar. 16, 2021, the entire content of which is incorporated herein by reference.
BACKGROUND OF THE INVENTION

1. Field of the Invention
(2) The present invention relates to an electronic device including a boost circuit.
2. Description of the Related Art
(3) In a case where input power is converted to boosted power by a boost circuit and the boosted power is used to operate a load, the load cannot be operated by the input power unless the input power is equal to or greater than self-consumed power of the boost circuit Thus, in an electronic device that includes a conventional boost circuit, a configuration has been proposed in which a storage capacitor is provided at an input of the boost circuit, power required to operate the load for a predetermined period of time is stored in the storage capacitor, and the load is intermittently operated by the power stored in the storage capacitor (for example, Japanese Patent Application Laid-open No. 2018-085888).
SUMMARY OF THE INVENTION
(4) An aspect of the present invention provides an electronic device including a boost circuit including a small number of storage capacitors or a boost circuit that uses a small storage capacitor having a small capacitance value.
(5) An electronic device including a boost circuit includes a boost circuit, an output circuit, a load, a first storage capacitor, a second storage capacitor, and an input terminal. The input terminal is connected to the first storage capacitor and the first boost circuit, the second storage capacitor is connected to the first boost circuit and the output circuit, and the load is connected to the output circuit.
(6) Since the second storage capacitor stores the boosted power boosted by the first boost circuit, even a small storage capacitor having a small capacitance value can store power for operating a load for a certain period of time. Since the first boost circuit can be operated a plurality of times to store the power of the first storage capacitor in the second storage capacitor, the first storage capacitor may also be a small storage capacitor having a capacitance value smaller than that of a storage capacitor having a conventional configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a circuit diagram illustrating an example of a first embodiment of the present invention.
(2) FIG. **2** is a circuit diagram illustrating an example of a first boost circuit of the first embodiment of the present invention.
(3) FIG. **3** is a circuit diagram illustrating an example of a first output circuit of the first embodiment of the present invention.
(4) FIG. **4** is a circuit diagram illustrating an example of a second output circuit of a second embodiment of the present invention.
(5) FIG. **5** is a circuit diagram illustrating an example of a third output circuit of a third embodiment of the present invention.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS
First Embodiment
(6) A first embodiment of the present invention will be described below with reference to the drawings. FIG. **1** is a circuit diagram illustrating an example of an electronic device **100** including a boost circuit according to the present embodiment.
(7) The electronic device **100** including the boost circuit according to the first embodiment of the present invention includes a first boost circuit **103**, a first output circuit **105**, a first storage capacitor **102**, a second storage capacitor **104**, a load **106**, a first input terminal **107**, and a second input terminal **108**.

(8) Connection of the electronic device **100** including the boost circuit will be described. An external power generation element **101** is externally connected between the first input terminal **107** and the second input terminal **108**. The first input terminal **107** is connected to a first terminal of the first storage capacitor **102** and a first terminal **151** of the first boost circuit **103**. The second input terminal **108** and a second terminal of the first storage capacitor **102** are connected to a GND power supply. A second terminal **152** of the first boost circuit **103** is connected to a first terminal of the second storage capacitor **104** and a first terminal **161** of the first output circuit **105**. A third terminal **153** of the first boost circuit **103** and a second terminal of the second storage capacitor **104** are connected to the GND power supply. A second terminal **162** of the first output circuit **105** is connected to a first terminal **171** of the load **106**. A third terminal **163** of the first output circuit **105** and a second terminal **172** of the load **106** are connected to the GND power supply.
(9) An operation of the electronic device **100** including the boost circuit of the first embodiment will be described. In response to the power generation element **101** outputting a first generated power, the first generated power is stored in the first storage capacitor **102**. In response to a stored voltage of the first storage capacitor **102** reaching an activation voltage of the first boost circuit **103**, the first boost circuit **103** is activated by stored power of the first storage capacitor **102**, and starts a boosting operation. In response to the first boost circuit **103** starting the boosting operation, the first boost circuit **103** converts the stored power of the first storage capacitor **102** to a first boosted power. The first boosted power is stored in the second storage capacitor **104**. In response to the stored voltage of the first storage capacitor **102** falling below an operating voltage of the first boost circuit **103**, the first boost circuit **103** stops the boosting operation. The generated power from the power generation element **101** is converted to the stored voltage of the first storage capacitor **102** again, and the same operation is repeated. In response to the stored voltage of the second storage capacitor **104** reaching an activation voltage of the first output circuit **105**, the first output circuit **105** converts a stored power of the second storage capacitor **104** to an output power. The output power is supplied to the load **106**, and the load **106** is operated by the output power.
(10) Here, for example, in a case where a capacitance value of the first storage capacitor **102** is 0.68 mF, a capacitance value of the second storage capacitor **104** is also 0.68 mF, an activation voltage of the first boost circuit **103** is 0.3 V, and an activation voltage of the first output circuit **105** is 1.9 V, in response to conversion of the stored power in the first storage capacitor **102** to the first boosted power being repeated approximately 100 times, the stored voltage of the second storage capacitor **104** reaches the activation voltage of the first output circuit **105**. The first output circuit **105** generates output power from the stored voltage of the second storage capacitor **104** and supplies the output power to the load **106**.
(11) As described above, the stored power of the first storage capacitor **102** is used as an activation power of the first boost circuit **103**. Thus, even in a case where the generated power of the power generation element **101** is smaller than the activation power of the first boost circuit **103**, the first boost circuit **103** can convert the stored power of the first storage capacitor **102** to the first boosted power. The stored power of the first storage capacitor **102** is smaller than a power for operating the load **106** for a predetermined period of time. However, the power for operating the load **106** for the predetermined period of time can be stored in the second storage capacitor **104** by activating the first boost circuit **103** a plurality of times. Further, the stored voltage of the second storage capacitor **104** is higher than the stored voltage of the first storage capacitor **102**. Thus, even in a case where the capacitance value of the second storage capacitor **104** is not increased, the stored power of the second storage capacitor **104** can operate the load **106** for the predetermined period of time.
(12) FIG. **2** illustrates an example of the first boost circuit **103**. The first boost circuit **103** includes a second boost circuit **202**, a first control circuit **208**, a third storage capacitor **205**, a fourth storage capacitor **207**, a first coil **201**, a first diode **203**, a first N-channel MOS transistor (hereinafter described as an “NMOS transistor”) **204**, a first switch **206**, the first terminal **151**, the second

terminal **152**, and the third terminal **153**.
(13) Connection of the first boost circuit **103** will be described. The first terminal **151** of the first boost circuit **103** is connected to a first terminal of the first coil **201** and a first terminal **250** of the second boost circuit **202**. A second terminal **251** of the second boost circuit **202** is connected to a control terminal of the first switch **206**. A third terminal **252** of the second boost circuit **202** is connected to a first terminal of the first switch **206** and a first terminal of the third storage capacitor **205**. A fourth terminal **253** of the second boost circuit **202** and a second terminal of the third storage capacitor **205** are connected to the GND power supply. A second terminal of the first switch **206** is connected to a first terminal of the fourth storage capacitor **207** and a second terminal **261** of the first control circuit **208**. A third terminal **262** of the first control circuit **208** and a second terminal of the fourth storage capacitor **207** are connected to the GND power supply. A drain terminal of the first NMOS transistor **204** is connected to a second terminal of the first coil **201** and an anode terminal of the first diode **203**. A gate terminal of the first NMOS transistor **204** is connected to a first terminal **260** of the first control circuit **208**. A source terminal of the first NMOS transistor **204** is connected to the GND power supply. A cathode terminal of the first diode **203** is connected to the second terminal **152** of the first boost circuit **103**. The third terminal **153** of the first boost circuit **103** is connected to the GND power supply.
(14) An operation of the first boost circuit **103** will be described. In response to a voltage of the first terminal **151** of the first boost circuit **103** reaching an activation voltage of the second boost circuit **202**, the second boost circuit **202** starts a boosting operation. In response to the second boost circuit **202** starting the boosting operation, the power supplied to the first terminal **151** of the first boost circuit **103** is converted to a second boosted power. The second boosted power is stored in the third storage capacitor **205**. The second boost circuit **202** monitors the stored voltage of the third storage capacitor **205** and detects that the stored power of the third storage capacitor **205** has been stored long enough for the first control circuit **208** to operate for a predetermined period of time. In response to the second boost circuit **202** detecting that the stored voltage of the third storage capacitor **205** has reached a predetermined voltage or greater, the second boost circuit **202** turns on the first switch **206** to supply the stored power of the third storage capacitor **205** to the second terminal **261** of the first control circuit **208**. The first control circuit **208** is operated by the stored power of the third storage capacitor **205**. From the first terminal **260**, the first control circuit **208** outputs a control signal for repeatedly turning on and off the first NMOS transistor **204**. The power stored in the first coil **201** while the first NMOS transistor **204** is on is output to the second terminal **152** of the first boost circuit **103** via the first diode **203** while the first NMOS transistor **204** is off, and thus the first boosted voltage is output from the second terminal **152** of the first boost circuit **103**.
(15) For example, in a case where a capacitance value of the third storage capacitor **205** is 1 μF, a capacitance value of the fourth storage capacitor **207** is 0.1 μF, and a stored voltage of the third storage capacitor **205** for turning on the first switch **206** is 2.4 V, the first control circuit **208** operates for approximately 0.1 seconds. Note that the fourth storage capacitor **207** is a smoothing capacitor for the power supply of the first control circuit **208**.
(16) As described above, the first control circuit **208** of the first boost circuit **103** operates for the predetermined period of time by using the stored power of the third storage capacitor **205**. Thus, the first boost circuit **103** can convert the power input to the first terminal **151** of the first boost circuit **103** to the first boosted power for the predetermined period of time and output the first boosted power from the second terminal **152** of the first boost circuit **103** for the predetermined period of time.
(17) FIG. **3** illustrates an example of the first output circuit **105**. The first output circuit **105** includes a first voltage detection circuit **306**, a second control circuit **308**, a second diode **301**, a third diode **309**, a fourth diode **310**, a fifth storage capacitor **303**, a sixth storage capacitor **304**, a seventh storage capacitor **311**, a second NMOS transistor **307**, a second switch **302**, a second coil

305, the first terminal **161**, the second terminal **162**, and the third terminal **163**. Here, the second control circuit **308**, the third diode **309**, the sixth storage capacitor **304**, the seventh storage capacitor **311**, the second NMOS transistor **307**, and the second coil **305** constitute a third boost circuit **356**.

(18) Connection of the first output circuit **105** will be described. The first terminal **161** of the first output circuit **105** is connected to an anode terminal of the second diode **301** and a first terminal of the second switch **302**. A cathode terminal of the second diode **301** is connected to a first terminal of the fifth storage capacitor **303**, a first terminal **350** of the first voltage detection circuit **306**, and a cathode terminal of the fourth diode **310**. A second terminal **351** of the first voltage detection circuit **306** is connected to a control terminal of the second switch **302**. A third terminal **352** of the first voltage detection circuit **306** and a second terminal of the fifth storage capacitor **303** are connected to the GND power supply. A second terminal of the second switch **302** is connected to a first terminal of the sixth storage capacitor **304** and a first terminal of the second coil **305**. A second terminal of the second coil **305** is connected to an anode terminal of the third diode **309** and a drain terminal of the second NMOS transistor **307**. A gate terminal of the second NMOS transistor **307** is connected to a first terminal **353** of the second control circuit **308**. A cathode terminal of the third diode **309** is connected to an anode terminal of the fourth diode **310**, a second terminal **354** of the second control circuit **308**, a first terminal of the seventh storage capacitor **311**, and the second terminal **162** of the first output circuit **105**. The third terminal **163** of the first output circuit **105**, a source terminal of the second NMOS transistor **307**, a third terminal **355** of the second control circuit **308**, a second terminal of the sixth storage capacitor **304**, and a second terminal of the seventh storage capacitor **311** are connected to the GND power supply.

(19) Operation of the first output circuit **105** illustrated in FIG. **3** will be described.

(20) The power supplied to the first terminal **161** of the first output circuit **105** is stored in the fifth storage capacitor **303** via the second diode **301**. The stored voltage of the fifth storage capacitor **303** is monitored by the first voltage detection circuit **306**. In response to the first voltage detection circuit **306** detecting that the stored voltage of the fifth storage capacitor **303** has reached a predetermined voltage, the first voltage detection circuit **306** turns on the second switch **302**. In response to the second switch **302** being turned on, the power supplied to the first terminal **161** of the first output circuit **105** is supplied to the second terminal **354** of the second control circuit **308** via the second coil **305** and the third diode **309**. The second control circuit **308** starts operating by using the power supplied to the second terminal **354**. From the first terminal **353**, the second control circuit **308** outputs a control signal for repeatedly turning on and off the second NMOS transistor **307**. The power stored in the second coil **305** while the second NMOS transistor **307** is on is output to the second terminal **162** of the first output circuit **105** via the third diode **309** while the second NMOS transistor **307** is off, and thus output power is output from the second terminal **162** of the first output circuit **105**. The output power is stored in the fifth storage capacitor **303** via the fourth diode **310**, while maintaining the operation of the second control circuit **308**.

(21) For example, in a case where a capacitance value of the fifth storage capacitor **303** is 10 μF, a detection voltage of the first voltage detection circuit **306** is 1.8 V, and a detection cancel voltage is 1.5 V, in response to a voltage of approximately 1.9 V being input to the first terminal **161** of the first output circuit **105**, the second switch **302** is turned on, and this activates the third boost circuit **356**. In response to the third boost circuit being activated, output power of, for example, 2 V is output. In response to the output power of 2 V being output, the fifth storage capacitor **303** is charged by the output power, and the voltage of the fifth storage capacitor **303** increases to 1.9 V without falling below 1.5 V. Thus, even in a case where the voltage at the first terminal **161** of the first output circuit **105** falls below 1.6 V, the second switch **302** can continue to be turned on. Note that the sixth storage capacitor **304** is a smoothing capacitor for the second coil **305**, and the seventh storage capacitor **311** is a smoothing capacitor for the power supply of the second control circuit **308**.

(22) As described above, in response to a predetermined voltage being input to the first terminal **161** of the first output circuit **105**, the first output circuit **105** can start the activation and output the output power. The load **106** is driven for a predetermined period of time by the output power of the first output circuit **105** supplied to the first terminal **171** of the load **106** from the second terminal **162** of the first output circuit **105**.
(23) As described above, according to the electronic device including the boost circuit of the present embodiment, by activating the first boost circuit **103** a plurality of times and storing power in the second storage capacitor **104**, the load **106** can be driven for the predetermined period of time by the power of the power generation element **101** by using the small first storage capacitor **102** and the small second storage capacitor **104**, both having a small capacitance value as compared with a conventional storage capacitor.
Second Embodiment
(24) A second embodiment of the present invention will be described below with reference to the drawings. The second embodiment has a configuration in which the first output circuit **105** of the first embodiment is replaced with a second output circuit **105***a*.
(25) FIG. **4** is a circuit diagram illustrating an example of the second output circuit **105***a* of the electronic device including the boost circuit according to the present embodiment. The second output circuit **105***a* includes the first voltage detection circuit **306**, the fifth storage capacitor **303**, the seventh storage capacitor **311**, the second switch **302**, the first terminal **161**, the second terminal **162**, and the third terminal **163**.
(26) Connection of the second output circuit **105***a* will be described. The first terminal **161** of the second output circuit **105***a* is connected to the first terminal of the fifth storage capacitor **303**, the first terminal **350** of the first voltage detection circuit **306**, and the first terminal of the second switch **302**. The second terminal **351** of the first voltage detection circuit **306** is connected to the control terminal of the second switch **302**. The third terminal **352** of the first voltage detection circuit **306** and the second terminal of the fifth storage capacitor **303** are connected to the GND power supply. The second terminal of the second switch **302** is connected to a first terminal of the seventh storage capacitor **311** and the second terminal **162** of the second output circuit **105***a*. The third terminal **163** of the second output circuit **105***a* and the second terminal of the seventh storage capacitor **311** are connected to the GND power supply.
(27) Operation of the second output circuit **105***a* of the electronic device including the boost circuit of the second embodiment will be described.
(28) The second output circuit **105***a* monitors a voltage of the power input to the first terminal **161** of the second output circuit **105***a* by the first voltage detection circuit **306**. In response to the first voltage detection circuit **306** detecting that the voltage at the first terminal **161** of the second output circuit **105***a* has reached a predetermined voltage, the first voltage detection circuit **306** turns on the second switch **302**. In response to the second switch **302** being turned on, the power input to the first terminal **161** of the second output circuit **105***a* is output as output power from the second terminal **162** of the second output circuit **105***a*. The load **106** is driven for a predetermined period of time by the output power of the second output circuit **105***a* supplied to the first terminal **171** of the load **106**. The seventh storage capacitor **311** is a smoothing capacitor for the power supply of the load **106**.
(29) For example, in a case where a detection voltage of the first voltage detection circuit **306** is 1.9 V and a detection cancel voltage is 1.6 V, in response to a voltage of 1.9 V being input to the first terminal **161** of the second output circuit **105***a*, the second output circuit **105***a* outputs the output power from the second terminal **162** of the second output circuit **105***a*. In response to the second output circuit **105***a* outputting the output power, the voltage at the first terminal **161** of the second output circuit **105***a* starts to decrease, and in response to the voltage at the first terminal **161** of the second output circuit **105***a* falling below 1.6 V, the second output circuit **105***a* stops outputting the output power. In response to a predetermined voltage being input to the first terminal **161** of the

second output circuit **105***a*, the second output circuit **105***a* can start the activation and drive the load **106** for the predetermined period of time.
(30) As described above, according to the electronic device including the boost circuit of the present embodiment, by activating the first boost circuit **103** a plurality of times and storing power in the second storage capacitor **104**, the load **106** can be driven for the predetermined period of time by the power of the power generation element **101** by using the small first storage capacitor **102** and the small second storage capacitor **104**, both having a small capacitance value as compared with a conventional storage capacitor.
Third Embodiment
(31) A third embodiment of the present invention will be described below with reference to the drawings. The third embodiment has a configuration in which the first output circuit **105** of the first embodiment is replaced with a third output circuit **105***b*.
(32) FIG. **5** is a circuit diagram illustrating an example of the third output circuit **105***b* of the electronic device including the boost circuit according to the present embodiment. The third output circuit **105***b* includes the first voltage detection circuit **306**, the second control circuit **308**, the third diode **309**, a fifth diode **313**, the seventh storage capacitor **311**, an eighth storage capacitor **316**, the second NMOS transistor **307**, a third NMOS transistor **315**, a first resistor **317**, the second coil **305**, the first terminal **161**, the second terminal **162**, and the third terminal **163**. Here, the second control circuit **308**, the third diode **309**, the seventh storage capacitor **311**, the second NMOS transistor **307**, and the second coil **305** constitute a fourth boost circuit **357**.
(33) Connection of the third output circuit **105***b* will be described. The first terminal **161** of the third output circuit **105***b* is connected to the first terminal of the second coil **305**. A second terminal of the second coil **305** is connected to an anode terminal of the third diode **309** and a drain terminal of the second NMOS transistor **307**. The cathode terminal of the third diode **309** is connected to the first terminal **350** of the first voltage detection circuit **306**, the second terminal **354** of the second control circuit **308**, the first terminal of the seventh storage capacitor **311**, and the second terminal **162** of the third output circuit **105***b*. The first terminal **353** of the second control circuit **308** is connected to the gate terminal of the second NMOS transistor **307**. The third terminal **355** of the second control circuit **308** is connected to the source terminal of the second NMOS transistor **307** and a drain terminal of the third NMOS transistor **315**.
(34) The second terminal **351** of the first voltage detection circuit **306** is connected to an anode terminal of the fifth diode **313**. The third terminal **352** of the first voltage detection circuit **306** is connected to the GND power supply. A cathode terminal of the fifth diode **313** is connected to a first terminal of the eighth storage capacitor **316**, a first terminal of the first resistor **317**, and a gate terminal of the third NMOS transistor **315**. The third terminal **163** of the third output circuit **105***b*, a second terminal of the eighth storage capacitor **316**, a second terminal of the first resistor **317**, and a source terminal of the third NMOS transistor **315** are connected to the GND power supply.
(35) Operation of the third output circuit **105***b* of the electronic device including the boost circuit of the third embodiment will be described. Power input to the first terminal **161** of the third output circuit **105***b* is supplied to the second terminal **162** of the third output circuit **105***b*, the second terminal **354** of the second control circuit **308**, and the first terminal **350** of the first voltage detection circuit **306** via the second coil **305** and the third diode **309**. The first voltage detection circuit **306** is operated by the power supplied to the first terminal **350** of the first voltage detection circuit **306**, and monitors the voltage at the first terminal **350** of the first voltage detection circuit **306**. In response to the first voltage detection circuit **306** detecting that the voltage at the first terminal **350** of the first voltage detection circuit **306** has reached a predetermined voltage or greater, the first voltage detection circuit **306** outputs the voltage at the first terminal **350** of the first voltage detection circuit **306** from the second terminal **351** of the first voltage detection circuit **306**. The voltage at the first terminal **350** of the first voltage detection circuit **306** output from the second terminal **351** of the first voltage detection circuit **306** is supplied to the gate terminal of the

third NMOS transistor **315** via the fifth diode **313**, and this turns on the third NMOS transistor **315**.
(36) In response to the third NMOS transistor **315** being turned on, the second control circuit **308** of the fourth boost circuit **357** starts operating. The second control circuit **308** repeats turning on and off the second NMOS transistor **307**. The power stored in the second coil **305** while the second NMOS transistor **307** is on is output to the second terminal **162** via the third diode **309** while the second NMOS transistor **307** is off, and thus output power of the fourth boost circuit **357** is output from the second terminal **162**. The voltage of the output power of the fourth boost circuit **357** is equal to or greater than the detection voltage of the first voltage detection circuit **306**. Thus, in response to the output power of the fourth boost circuit **357** being supplied to the first terminal **350** of the first voltage detection circuit **306**, the first voltage detection circuit **306** outputs the voltage at the first terminal **350** of the first voltage detection circuit **306** from the second terminal **351** of the first voltage detection circuit **306**. Thus, the third NMOS transistor **315** continues to be turned on. Note that the seventh storage capacitor **311** is a smoothing capacitor for the power supply of the second control circuit **308**, and the eighth storage capacitor **316** is a capacitor for holding the gate voltage of the third NMOS transistor **315** for a predetermined period of time even in a case where no voltage is output from the second terminal **351** of the first voltage detection circuit **306**. Further, the first resistor **317** is a resistor for discharging the seventh storage capacitor **311**.
(37) For example, in a case where a detection voltage of the first voltage detection circuit **306** is 1.8 V, a detection cancel voltage is 1.7 V, a capacitance value of the eighth storage capacitor **316** is 1 μF, a resistance value of the first resistor **317** is 1 MΩ, and a voltage of the output power of the fourth boost circuit **357** is 2.0 V, in response to a voltage of 1.9 V being input to the first terminal **161** of the third output circuit **105***b*, the first voltage detection circuit **306** of the third output circuit **105***b* outputs the voltage at the first terminal **350** of the first voltage detection circuit **306** from the second terminal **351** of the first voltage detection circuit **306**. Thus, the second control circuit **308** starts operating.
(38) In response to the second control circuit **308** starting to operate, since the voltage of the input power falls below 1.8 V, the first voltage detection circuit **306** does not output the voltage at the first terminal **350** of the first voltage detection circuit **306** from the second terminal **351** of the first voltage detection circuit **306**. At this time, since the voltage of the third NMOS transistor **315** hardly decreases due to the fifth diode **313** and the eighth storage capacitor **316**, the third NMOS transistor **315** can maintain the on state. As a result, the second control circuit **308** can maintain the operation, and output power of 2.0 V is output from the fourth boost circuit **357**. In response to output power of 2.0 V being output from the fourth boost circuit **357**, the first voltage detection circuit **306** outputs the voltage at the first terminal **350** of the first voltage detection circuit **306** from the second terminal **351** of the first voltage detection circuit **306**. Thus, the fourth boost circuit **357** can continue to output the output power from the second terminal **162** of the third output circuit **105***b* until the voltage of the power input to the first terminal **161** of the third output circuit **105***b* drops to approximately 0.1 V.
(39) As described above, in response to a predetermined voltage being input to the first terminal **161**, the third output circuit **105***b* can start activation and drive the load **106** for a predetermined period of time.
(40) As described above, according to the electronic device including the boost circuit of the present embodiment, by activating the first boost circuit **103** a plurality of times and storing power in the second storage capacitor **104**, the load **106** can be driven for the predetermined period of time using the power of the power generation element by using the small first storage capacitor **102** and the small second storage capacitor **104**, both having a small capacitance value as compared with a conventional storage capacitor.
(41) According to the present invention, the first boost circuit can be operated by the stored power of the small first storage capacitor having a small capacitance value and that is charged by the power of the power generation element, and the power of the first storage capacitor can be stored in

the second storage capacitor. Since the stored power stored in the second storage capacitor has a higher voltage than that of the stored power stored in the first storage capacitor, the power used for operating the load for the predetermined period of time can be stored by the small second storage capacitor having a small capacitance value. In response to the first boost circuit operating a plurality of times and the power for operating the load for the predetermined period of time being stored in the second storage capacitor, the output circuit outputs the power for operating the load for the predetermined period of time to the load.

(42) Since the second storage capacitor of the present invention stores the power for operating the load for the predetermined period of time at a further higher voltage, a small storage capacitor having a small capacitance value may be used. Further, since the first storage capacitor of the present invention, which is directly connected to the power generation element, operates the first boost circuit a plurality of times (for example, 100 times in the first embodiment) to store predetermined power in the second storage capacitor, a small storage capacitor having a capacitance value smaller than that of an electronic device including a conventional boost circuit may be used.

Claims

1. An electronic device comprising: an input terminal connected to a power generator that supplies input electricity; a first storage capacitor configured to store the input electricity therein supplied through the input terminal, wherein the first storage capacitor is operable to repeat charge and discharge of the input electricity stored therein; a first boost circuit activated to become operative, from being inactive, to start to elevate the input electricity stored in the first storage capacitor to first elevated electricity when the input electricity stored in the first storage capacitor increases its voltage to a first predetermined voltage during charging the first storage capacitor with the input electricity and deactivated to become inactive when the input electricity stored in the first storage capacitor is discharged through boosting operation of the first boost circuit; a second storage capacitor configured to store the first elevated electricity therein supplied from the first boost circuit, wherein the second storage capacitor is operable to repeat charge and discharge of the first elevated electricity stored therein; a first output circuit activated to become operative, from being inactive, to start to elevate the first elevated electricity stored in the second storage capacitor to second elevated electricity when the first elevated electricity stored in the second storage capacitor increases its voltage to a second predetermined voltage during charging the second storage capacitor with the first elevated electricity and deactivated to become inactive when the first elevated electricity stored in the second storage capacitor is discharged through boosting operation of the first output circuit; and a load connected to the first output circuit to receive the second elevated electricity from the first output circuit.

2. The electronic device according to claim 1, wherein the first elevated electricity stored in the second storage capacitor has a voltage higher than a voltage of the input electricity stored in the first storage capacitor.

3. The electronic device according to claim 2, wherein the first output circuit includes a first voltage detection circuit, a third boost circuit, and a second switch, the second switch is disposed between an input terminal of the first output circuit and the third boost circuit, and a control terminal of the second switch is connected to the first voltage detection circuit.

4. The electronic device according to claim 2, wherein the first output circuit includes a first voltage detection circuit and a second switch, the second switch is disposed between an input terminal and an output terminal of the first output circuit, and a control terminal of the second switch is connected to the first voltage detection circuit.

5. The electronic device according to claim 2, wherein the first output circuit includes a first voltage detection circuit, a fourth boost circuit, and a transistor switch, the transistor switch is

disposed between a power supply terminal of the fourth boost circuit and a GND power supply, and a control terminal of the transistor switch is connected to the first voltage detection circuit.
6. The electronic device according to claim 1, wherein the first output circuit includes a first voltage detection circuit, a third boost circuit, and a second switch, the second switch is disposed between an input terminal of the first output circuit and the third boost circuit, and a control terminal of the second switch is connected to the first voltage detection circuit.
7. The electronic device according to claim 1, wherein the first output circuit includes a first voltage detection circuit and a second switch, the second switch is disposed between an input terminal and an output terminal of the first output circuit, and a control terminal of the second switch is connected to the first voltage detection circuit.
8. The electronic device according to claim 1, wherein the first output circuit includes a first voltage detection circuit, a fourth boost circuit, and a transistor switch, the transistor switch is disposed between a power supply terminal of the fourth boost circuit and a GND power supply, and a control terminal of the transistor switch is connected to the first voltage detection circuit.
9. The electronic device according to claim 1, wherein the first boost circuit comprises: a second boost circuit activated to elevate the input electricity stored in the first storage capacitor and output a third elevated electricity; a third storage capacitor configured to store the third elevated electricity therein supplied from the second boost circuit; a first control circuit activated by a stored voltage of the third storage capacitor to become operative, from being inactive, to start to output an on/off signal to inductively elevate the input electricity stored in the first storage capacitor to the first elevated electricity; and a first switch turned on to connect the third storage capacitor to the first control circuit to activate the first control circuit, when the stored voltage of the third storage capacitor increases to a third predetermined voltage during charging the third storage capacitor with the third elevated electricity.
10. The electronic device according to claim 9, wherein the first boost circuit further comprises a fourth storage capacitor configured to smooth power supply to the first control circuit.
11. The electronic device according to claim 1, wherein the first output circuit comprises: a third boost circuit activated to elevate the first elevated electricity stored in the second storage capacitor to the second elevated electricity; a fifth storage capacitor configured to store the first elevated electricity therein supplied from the second storage capacitor; a second switch turned on to connect the second storage capacitor to the third boost circuit to activate the third boost circuit; a second control circuit included in the third boost circuit and activated by a stored voltage of the fifth storage capacitor to start to output an on/off signal to inductively elevate the first elevated electricity stored in the second storage capacitor to the second elevated electricity; and a first voltage detection circuit configured to monitor the stored voltage of the fifth storage capacitor and turn on the second switch to activate the second control circuit when the stored voltage of the fifth storage capacitor increases to a fourth predetermined voltage during charging the fifth storage capacitor with the first elevated electricity.
12. The electronic device according to claim 11, wherein the first output circuit is configured to supply the second elevated electricity back to the first voltage detection circuit to keep the second switch turned on and back to the second control circuit to keep the second control circuit activated.
13. The electronic device according to claim 11, wherein the third boost circuit further comprises a sixth storage capacitor configured to smooth power supply to the third boost circuit and a seventh storage capacitor configured to smooth power supply to the load.
14. The electronic device according to claim 1, wherein the first output circuit comprises: a fifth storage capacitor configured to store the first elevated electricity therein supplied from the second storage capacitor; a second switch turned on to connect the second storage capacitor directly to the load to supply the first elevated electricity stored in the second storage capacitor directly to the load; and a first voltage detection circuit configured to monitor a stored voltage of the fifth storage capacitor and turn on the second switch when the stored voltage of the fifth storage capacitor

increases to a fifth predetermined voltage during charging the fifth storage capacitor with the first elevated electricity.

15. The electronic device according to claim 1, wherein the first output circuit comprises: a fourth boost circuit activated to elevate the first elevated electricity stored in the second storage capacitor to the second elevated electricity; a second control circuit included in the fourth boost circuit and activated by the second elevated electricity to start to output an on/off signal to inductively elevate the first elevated electricity stored in the second storage capacitor to the second elevated electricity; a transistor switch turned on to make operable the on/off signal from the second control circuit to enable elevation of the first elevated electricity to the second elevated electricity and turned off to make inoperative the on/off signal from the second control circuit to disable elevation of the first elevated electricity to the second elevated electricity; and a first voltage detection circuit configured to monitor the second elevated electricity and turn on the transistor switch to enable elevation of the first elevated electricity to the second elevated electricity, when the second elevated electricity increases to a fifth predetermined level.

16. The electronic device according to claim 15, wherein the fourth boost circuit further comprises a seventh storage capacitor configured to smooth power supply to the load.
